Identify the decision alternatives, the states of nature, the uncertain event, and the payoff components. Then construct the corresponding payoff table.

Context Descriptions

1. A school event company must choose between a compact snack stand and a full snack stand before attendance is known. Attendance may be high or low. Analysts have already converted each possible consequence into direct profit measured in thousand USD. The compact stand produces a profit of 18 thousand USD under high attendance and 11 thousand USD under low attendance. The full stand produces a profit of 27 thousand USD under high attendance and 6 thousand USD under low attendance.
2. A weekend pop-up business must choose between an indoor booth and a mobile stall before market demand is known. Demand may be strong or weak. The indoor booth contributes a fixed 550 thousand JPY to profit, while the mobile stall contributes a fixed 420 thousand JPY. Strong demand adds a fixed market bonus of 180 thousand JPY to either alternative. Weak demand instead reduces profit by a fixed 90 thousand JPY because of permit and cleanup expenses.
3. A local delivery firm must choose between a bicycle fleet and an electric-van fleet before traffic conditions are known. Traffic may be clear or congested. The bicycle fleet earns 4 thousand CNY per completed delivery batch and is expected to complete 14 delivery batches in clear traffic and 10 in congested traffic. The electric-van fleet earns 6 thousand CNY per completed delivery batch and is expected to complete 12 delivery batches in clear traffic and 8 in congested traffic. Clear traffic also adds a fixed scheduling bonus of 5 thousand CNY to either fleet, while congestion creates a fixed loss of 4 thousand CNY.
4. A city-tour operator must choose between a standard tour and a premium tour before tourism demand is known. Demand may be high or low. The standard tour contributes 2 thousand EUR per group booking and also provides a fixed contract contribution of 6 thousand EUR. The premium tour contributes 4 thousand EUR per group booking and provides a fixed contract contribution of 10 thousand EUR. High tourism demand adds another 3 thousand EUR per group booking to either tour, while low demand adds 1 thousand EUR per group booking. The standard tour is expected to complete 16 group bookings under high demand and 10 under low demand; the premium tour is expected to complete 12 group bookings under high demand and 7 under low demand.
5. A solar-installation company must choose between a standard installation crew and an expanded installation crew before customer orders are known. Orders may be strong, regular, or weak. The standard crew contributes 3 thousand GBP per completed installation and provides a fixed contract contribution of 8 thousand GBP. The expanded crew contributes 4 thousand GBP per completed installation and provides a fixed contract contribution of 5 thousand GBP. Strong orders add 2 thousand GBP per completed installation and a fixed scheduling bonus of 6 thousand GBP. Regular orders add 1 thousand GBP per completed installation and no fixed adjustment. Weak orders reduce the contribution by 1 thousand GBP per completed installation and create a fixed idle-time loss of 4 thousand GBP. The standard crew is expected to complete 10 installations under strong orders, 7 under regular orders, and 4 under weak orders. The expanded crew is expected to complete 14, 9, and 5 installations under the same respective states.
6. A coffee exporter must choose among contracted storage, a rented warehouse, and an automated distribution hub before export demand and shipping conditions are known. Conditions may be favorable, normal, or adverse. Contracted storage contributes 4 million COP per export lot handled but requires a fixed storage commitment of 12 million COP. The rented warehouse contributes 5 million COP per export lot but requires a fixed commitment of 20 million COP. The automated hub contributes 6 million COP per export lot but requires a fixed commitment of 32 million COP. Favorable conditions add 2 million COP per export lot and a fixed logistics bonus of 8 million COP. Normal conditions make no additional per-lot or fixed adjustment. Adverse conditions reduce the contribution by 2 million COP per export lot and create a fixed disruption loss of 10 million COP. Contracted storage is expected to handle 18 export lots under favorable conditions, 13 under normal conditions, and 8 under adverse conditions. The rented warehouse is expected to handle 22, 15, and 9 export lots, while the automated hub is expected to handle 26, 17, and 10 export lots under the same respective states.

1 Solutions: Extraction

Problems:

1.1 Exercise 1: School event snack stand

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Compact stand	**A2:** Full stand
Statesofnature	**S1:** High attendance	**S2:** Low attendance
Events	Attendance level after the stand is selected	
Consequences	Profit in thousand USD for each stand–attendance pair	

C2 - Building the payoff table. Relevant values. The context supplies a direct profit for every stand–attendance pair.

Abstract component	Context component	Math symbol	Corresponding value
Direct payoff	Profit stated for each stand under each attendance level	$P_{ay}(A_i, S_j)$	Compact–high: 18 thousand USD; Compact–low: 11 thousand USD; Full–high: 27 thousand USD; Full–low: 6 thousand USD

Payoff equation. Because these are direct-payoff data, $P_{ay}(A_i, S_j)$ is read directly from the context.

Calculations.

Stand	High attendance	Low attendance
Compact stand	Direct context value: 18	Direct context value: 11
Full stand	Direct context value: 27	Direct context value: 6

Final payoff table (profit in thousand USD).

Stand	High attendance	Low attendance
Compact stand	18	11
Full stand	27	6

1.2 Exercise 2: Weekend pop-up business

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Indoor booth	**A2:** Mobile stall
Statesofnature	**S1:** Strong demand	**S2:** Weak demand
Events	Market demand after the selling format is selected	
Consequences	Profit in thousand JPY for each format–demand pair	

C2 - Building the payoff table. Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Fixed, alternative dependent	Fixed contribution to profit from the chosen selling format	$f^A(A_i)$	Indoor booth: 550 thousand JPY; Mobile stall: 420 thousand JPY
Fixed, state dependent	Market bonus under strong demand or permit-and-cleanup loss under weak demand	$f^S(S_j)$	Strong demand: 180 thousand JPY; Weak demand: −90 thousand JPY

Payoff equation.

$$P_{ay}(A_i, S_j) = f^A(A_i) + f^S(S_j).$$

Calculations (thousand JPY).

Format	Strong demand	Weak demand
Indoor booth	$550 + 180 = 730$	$550 + (-90) = 460$
Mobile stall	$420 + 180 = 600$	$420 + (-90) = 330$

Final payoff table (profit in thousand JPY).

Format	Strong demand	Weak demand
Indoor booth	730	460
Mobile stall	600	330

1.3 Exercise 3: Local delivery fleet

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Bicycle fleet	**A2:** Electric-van fleet
Statesofnature	**S1:** Clear traffic	**S2:** Congested traffic
Events	Traffic conditions after the fleet is selected	
Consequences	Profit in thousand CNY for each fleet–traffic pair	

C2 - Building the payoff table. Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Quantity, alternative–state dependent	Expected completed delivery batches for each fleet under each traffic condition	$q(A_i, S_j)$	Bicycle–clear: 14 batches; Bicycle–congested: 10 batches; Electric van–clear: 12 batches; Electric van–congested: 8 batches
Variable, alternative dependent	Earnings per completed delivery batch for the chosen fleet	$v^A(A_i)$	Bicycle fleet: 4 thousand CNY/batch; Electric-van fleet: 6 thousand CNY/batch
Fixed, state dependent	Scheduling bonus in clear traffic or congestion loss	$f^S(S_j)$	Clear traffic: 5 thousand CNY; Congested traffic: -4 thousand CNY

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v^A(A_i) + f^S(S_j).$$

Calculations (thousand CNY).

Fleet	Clear traffic	Congested traffic
Bicycle fleet	$14(4) + 5 = 61$	$10(4) + (-4) = 36$
Electric-van fleet	$12(6) + 5 = 77$	$8(6) + (-4) = 44$

Final payoff table (profit in thousand CNY).

Fleet	Clear traffic	Congested traffic
Bicycle fleet	61	36
Electric-van fleet	77	44

1.4 Exercise 4: City-tour format

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Standard tour	**A2:** Premium tour
Statesofnature	**S1:** High tourism demand	**S2:** Low tourism demand
Events	Tourism demand after the tour format is selected	
Consequences	Profit in thousand EUR for each tour–demand pair	

C2 - Building the payoff table. Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Quantity, alternative–state dependent	Expected group bookings for each tour under each tourism-demand level	$q(A_i, S_j)$	Standard–high: 16 bookings; Standard–low: 10 bookings; Premium–high: 12 bookings; Premium–low: 7 bookings
Variable, alternative dependent	Contribution per group booking from the chosen tour	$v^A(A_i)$	Standard tour: 2 thousand EUR/booking; Premium tour: 4 thousand EUR/booking
Variable, state dependent	Additional contribution per group booking from tourism demand	$v^S(S_j)$	High demand: 3 thousand EUR/booking; Low demand: 1 thousand EUR/booking
Fixed, alternative dependent	Fixed contract contribution from the chosen tour	$f^A(A_i)$	Standard tour: 6 thousand EUR; Premium tour: 10 thousand EUR

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\left[v^A(A_i) + v^S(S_j)\right] + f^A(A_i).$$

Calculations (thousand EUR).

Tour	High tourism demand	Low tourism demand
Standard tour	$16(2+3)+6=86$	$10(2+1)+6=36$
Premium tour	$12(4+3)+10=94$	$7(4+1)+10=45$

Final payoff table (profit in thousand EUR).

Tour	High tourism demand	Low tourism demand
Standard tour	86	36
Premium tour	94	45

1.5 Exercise 5: Solar-installation crew

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Standard installation crew	**A2:** Expanded installation crew	
Statesofnature	**S1:** Strong orders	**S2:** Regular orders	**S3:** Weak orders
Events	Customer-order level after the crew size is selected		
Consequences	Profit in thousand GBP for each crew–order pair		

C2 - Building the payoff table. Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Quantity, alternative–state dependent	Expected completed installations for each crew under each order level	$q(A_i, S_j)$	Standard–strong: 10; Standard–regular: 7; Standard–weak: 4 installations; Expanded–strong: 14; Expanded–regular: 9; Expanded–weak: 5 installations
Variable, alternative dependent	Contribution per completed installation from the chosen crew	$v^A(A_i)$	Standard crew: 3 thousand GBP/installation; Expanded crew: 4 thousand GBP/installation
Variable, state dependent	Per-installation addition or reduction from the order level	$v^S(S_j)$	Strong: 2 thousand GBP/installation; Regular: 1 thousand GBP/installation; Weak: -1 thousand GBP/installation
Fixed, alternative dependent	Fixed contract contribution from the chosen crew	$f^A(A_i)$	Standard crew: 8 thousand GBP; Expanded crew: 5 thousand GBP
Fixed, state dependent	Scheduling bonus, no regular-order adjustment, or idle-time loss	$f^S(S_j)$	Strong: 6 thousand GBP; Regular: 0 thousand GBP; Weak: -4 thousand GBP

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\left[v^A(A_i) + v^S(S_j)\right] + f^A(A_i) + f^S(S_j).$$

Calculations (thousand GBP).

Crew	Strong orders	Regular orders	Weak orders
Standard installation crew	$10(3+2)+8+6=64$	$7(3+1)+8+0=36$	$4(3+(-1))+8+(-4)=12$
Expanded installation crew	$14(4+2)+5+6=95$	$9(4+1)+5+0=50$	$5(4+(-1))+5+(-4)=16$

Final payoff table (profit in thousand GBP).

Crew	Strong orders	Regular orders	Weak orders
Standard installation crew	64	36	12
Expanded installation crew	95	50	16

1.6 Exercise 6: Coffee-export storage strategy

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Contracted storage	**A2:** Rented warehouse	**A3:** Automated distribution hub
Statesofnature	**S1:** Favorable conditions	**S2:** Normal conditions	**S3:** Adverse conditions
Events	Export-demand and shipping conditions after the storage strategy is selected		
Consequences	Profit in million COP for each storage–condition pair		

C2 - Building the payoff table. Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Quantity, alternative–state dependent	Expected export lots handled by each storage strategy under each condition	$q(A_i, S_j)$	Contracted–favorable: 18; normal: 13; adverse: 8 lots; Rented–favorable: 22; normal: 15; adverse: 9 lots; Automated–favorable: 26; normal: 17; adverse: 10 lots
Variable, alternative dependent	Contribution per export lot from the chosen storage strategy	$v^A(A_i)$	Contracted: 4 million COP/lot; Rented: 5 million COP/lot; Automated: 6 million COP/lot
Variable, state dependent	Per-lot addition, no adjustment, or reduction from conditions	$v^S(S_j)$	Favorable: 2 million COP/lot; Normal: 0 million COP/lot; Adverse: -2 million COP/lot
Fixed, alternative dependent	Fixed storage commitment required by the chosen strategy	$f^A(A_i)$	Contracted: -12 million COP; Rented: -20 million COP; Automated: -32 million COP
Fixed, state dependent	Logistics bonus, no adjustment, or disruption loss from conditions	$f^S(S_j)$	Favorable: 8 million COP; Normal: 0 million COP; Adverse: -10 million COP

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\left[v^A(A_i) + v^S(S_j)\right] + f^A(A_i) + f^S(S_j).$$

Calculations (million COP).

Storage strategy	Favorable	Normal	Adverse
Contracted storage	$18(4+2)+(-12)+8=104$	$13(4+0)+(-12)+0=40$	$8(4+(-2))+(-12)+(-10)=-6$
Rented warehouse	$22(5+2)+(-20)+8=142$	$15(5+0)+(-20)+0=55$	$9(5+(-2))+(-20)+(-10)=-3$
Automated distribution hub	$26(6+2)+(-32)+8=184$	$17(6+0)+(-32)+0=70$	$10(6+(-2))+(-32)+(-10)=-2$

Final payoff table (profit in million COP).

Storage strategy	Favorable	Normal	Adverse
Contracted storage	104	40	-6
Rented warehouse	142	55	-3
Automated distribution hub	184	70	-2